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SHORT COMMUNICATION

The utility of ChatGPT as a generative medical translator

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Abstract

Purpose Large language models continue to dramatically change the medical landscape. We aimed to explore the utility of ChatGPT in providing accurate, actionable, and understandable generative medical translations in English, Spanish, and Mandarin pertaining to Otolaryngology.

Methods Responses of GPT-4 to commonly asked patient questions listed on official otolaryngology clinical practice guidelines (CPG) were evaluated with the Patient Education materials Assessment Tool-printable (PEMAT-P.) Additional critical elements were identified a priori to evaluate ChatGPT's accuracy and thoroughness in its responses. Multiple fluent speakers of English, Mandarin, and Spanish evaluated each response generated by ChatGPT.

Results Total PEMAT-P scores differed between English, Mandarin, and Spanish GPT-4 generated responses depicting a moderate effect size of language, Eta-Square 0.07 with scores ranging from 73 to 77 (P-value = 0.03). Overall understandability scores did not differ between English, Mandarin, and Spanish depicting a small effect size of language, Eta-Square 0.02 scores ranging from 76 to 79 (P-value = 0.17), nor did overall actionability scores Eta-Square 0 score ranging 66–73 (P-value = 0.44). Overall a priori procedure-specific responses similarly did not differ between English, Spanish, and Mandarin Eta-Square 0.02 scores ranging 61–78 (P-value = 0.22).

Conclusion GPT-4 produces accurate, understandable, and actionable outputs in English, Spanish, and Mandarin. Responses generated by GPT-4 in Spanish and Mandarin are comparable to English counterparts indicating a novel use for these models within Otolaryngology, and implications for bridging healthcare access and literacy gaps.

Level of evidence IV.

Keywords Large language models · Artificial intelligence · Disparities

Introduction

The advent of generative pretrained transformers (GPT), interactive chatbots, and large language models (LLMs) continues to rapidly shift the medical and research landscape. Despite immense academic and industry interest in these technological platforms, functional healthcare use cases of these technologies at present still lack sufficient objective assessment in the literature, though has grown at an unprecedented rate [1–7]. Previous work has demonstrated ChatGPT's ability to address potential healthcare gaps by providing accurate, understandable, and actionable discharge instructions at a reading level that is appropriate and adaptable to its user [8]. Similar studies have depicted its ability to aid in medical knowledge acquisition and medical recommendations [9–13].

Language barriers represent a social determinant of health with significant impact on outcomes, adverse events, patient and provider satisfaction, healthcare utilization, and costs [14]. The ability to utilize ChatGPT, however, to improve translation of medical services represents a potential partial remedy for this source of health inequity. Studies focused on ChatGPT's ability to respond in over ninety-five languages appear to be underleveraged, though continue to foster interest in expanding its use cases in the clinical setting to those

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of different backgrounds with diverse language requirements [15–17]. This qualitative study further assesses the utility of ChatGPT (GPT-4) to improve health literacy and bridge language disparities by providing accurate, understandable, and actionable answers to common otolaryngology medical questions in English, Spanish, and Mandarin.

Materials and methods

We analyzed the responses of GPT-4 to commonly asked patient questions listed on official otolaryngology clinical practice guidelines (CPG) for Otology, Head & Neck, Laryngology, Rhinology, Pediatrics, Facial Plastics and Reconstructive Surgery, and Pain Management [18]. Stanford University Institutional Review Board deemed this study exempt from review and waived the informed consent requirement. SRQR reporting guidelines were followed.

Select FAQs from CPGs were obtained and provided to ChatGPT (GPT-4.) Questions followed the format "Are there any risks related to tonsillectomy. Respond in "___" where _____ was replaced with English, Spanish, or Mandarin (Supplement 1). These three languages were chosen as a pilot study to evaluate ChatGPT's ability to translate more widely spoken languages. Results were extracted and blinded by DRG, then scored by KH, LL, & OG. Individuals translating and scoring non-English responses were certified by Stanford Translational Services to engage with patients in said language or had equivalent native proficiency. At least two certified translators or native speakers scored each response, with differences reconciled by agreement or third-party arbitration.

The primary outcome was the Patient Education materials Assessment Tool-printable (PEMAT-P) [19]. This validated outcome metric evaluates printed materials understandability and actionability for patients with differing levels of healthcare literacy. The PEMAT-P uses 17 questions to assess understandability and 7 for actionability, for a total of 24 questions. The reviewer must determine whether the material achieves specific educational objectives, and each question is scored in a binary fashion: 1 (agree) or 0 (disagree). The total, understandability, and actionability scores are each calculated by summing each individual question score and dividing by the total possible score.

To evaluate the clinical accuracy and thoroughness of ChatGPT's responses, three additional critical response elements were identified a priori for each procedure by two otolaryngologists (NFA and YJL). These presence of these details within ChatGPT's responses was used to evaluate whether the responses to CPG questions generated by ChatGPT were comprehensive, valid, and accurate. The presence or absence of these procedure-specific items from GPT-4 responses were used as secondary outcome measures. For example, CPG questions in Otology included "What is sudden hearing loss?" Critical elements in generated responses looked for appropriate clinical definitions, descriptions of types of hearing loss, and mentions of potential causes (Supplemental 1).

Scores were compared using 1-way analysis of variance and Kruskal–Wallis with N2 (90% CI) as the appropriate effect size [20].

Results

For individual CPG specialty related questions, ChatGPT generated English PEMAT-P total scores ranged from 76 to 79, understandability scores ranged from 77 to 80, actionability ranged 67–75, and procedure-specific item scores ranged 67–89. For Mandarin, PEMAT-P total scores ranged from 69–79, understandability scores ranged 70–83, actionability ranged 50–75, and procedure specific item scores ranged 33–89. For Spanish, PEMAT-P total scores ranged from 67 to 79, understandability scores ranged 73–83, actionability scores ranged 42–75, and procedure specific item scores ranged 33–78 (Table 1).

Overall total PEMAT-P scores differed between English, Mandarin, and Spanish GPT-4 generated responses Eta-Square 0.07 scores ranging from 73 to 77 (P-value = 0.03). Overall understandability scores did not differ between English, Mandarin, and Spanish Eta-Square 0.02 scores ranging from 76 to 79 (P-value = 0.17), nor did overall actionability scores Eta-Square 0 score ranging 66–73 (P-value = 0.44). A priori procedure-specific responses similarly did not differ between English, Spanish, and Mandarin Eta-Square 0.02 scores ranging 61–78 (P-value = 0.22) (Table 2).

Discussion

Language barriers are a known contributor to miscommunication, reduced patient and provider satisfaction, worse quality of care and higher rates of adverse events [21–23]. They can also lead to increased healthcare costs, longer clinic appointments, and greater utilization of healthcare services [22, 24]. The use of interpreters has been shown to improve patient outcomes [21, 25]. The release of ChatGPT -4 and other highly proficient, publicly available LLMs into society represents a landmark moment for the democratization of education and patient access to medical information. Increased access to a resource that can reliably and accurately translate medical information into other languages could potentially alleviate some of the issues patients and providers face with language barriers.

Our findings suggest that ChatGPT (GPT-4) can translate responses to frequently asked, patient-focused

Table 1 Understandability, actionability, and procedure-specific scores for each clinical practice guideline (CPG)

Procedure and CPG	PEMAT-P total score, %	PEMAT-P understandability score, %	PEMAT-P actionability score, %	Procedure specific item score, %
Otology CPG				
English	76	77	75	89
Mandarin	69	77	50	89
Spanish	67	77	42	78
H&N CPG				
English	76	80	67	67
Mandarin	71	70	75	56
Spanish	74	73	75	56
Laryngology CPG				
English	76	80	67	78
Mandarin	76	80	67	33
Spanish	79	80	75	33
Rhinology CPG				
English	79	80	75	78
Mandarin	76	80	67	56
Spanish	79	80	75	67
Pediatrics CPG				
English	79	80	75	78
Mandarin	69	77	50	67
Spanish	76	80	67	56
Facial plastics CPG				
English	76	77	75	78
Mandarin	79	83	67	78
Spanish	79	83	67	78
Pain management CPG				
English	79	80	75	NA
Mandarin	73	72	75	NA
Spanish	76	77	75	NA

Table 2 Comparison of GPT-4 responses total, understandability, actionability, and specialty specific questions in English, Mandarin, and Spanish

	English	Mandarin	Spanish	P-value	Eta-Squared (90% CI)	Stats test
PEMAT-P total	0.77 ± 0.03	0.73 ± 0.06	0.76 ± 0.06	0.03	0.07 (0.001–0.23)	Kruskal–Wallis
PEMAT-P understandability	0.79 ± 0.03	0.76 ± 0.06	0.78 ± 0.07	0.17	0.02 (0–0.16)	Kruskal–Wallis
PEMAT-P actionability	0.73 ± 0.07	0.66 ± 0.22	0.69 ± 0.17	0.44	0 (0–0.09)	Kruskal–Wallis
Procedure-specific questions	0.78 ± 0.23	0.63 ± 0.34	0.61 ± 0.31	0.22	0.02 (0–0.18)	Kruskal–Wallis

otolaryngology questions into multiple languages in an accurate, understandable, and actionable manner. While our findings suggest that language may impart a medium effect size on overall PEMAT-P scores, minimal differences are seen between responses in English, Spanish, and Mandarin with regards to actionability and understandability. In situations where access to language specific resources is limited, ChatGPT may serve as an adjunct to bridge healthcare gaps from a provider standpoint, as well as bridge healthcare gaps from patients of diverse language and healthcare literacy backgrounds [26].

This study serves to answer a critical question regarding the applicability of LLMs in a medical context to languages other than English, the language which these models have historically been primarily trained. While initial findings are promising, further development of these models is necessary to promote equity in their ability and application.

Study limitations include the narrow range of questions provided from Otolaryngology CPGs, and the limited selection of languages (English, Spanish & Mandarin) that contain strong underlying training datasets that form the basis of GPT-4. GPT-4 limitations include training data up to September 2021, limited access to recent findings or changes in the medical literature after that point, as well as limitations based on the technical literacy of the individuals interacting with GPT-4.

Conclusion

GPT-4 produces accurate, understandable, and actionable outputs in English, Spanish, and Mandarin. Outputs in Spanish and Mandarin are comparable to English counterparts indicating a novel use for these models within Otolaryngology, and implications for bridging healthcare access and literacy gaps.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00405-024-08708-8.

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Declarations

Conflict of interest None of the authors have any financial disclosures or conflicts of interest relevant to this manuscript.

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